

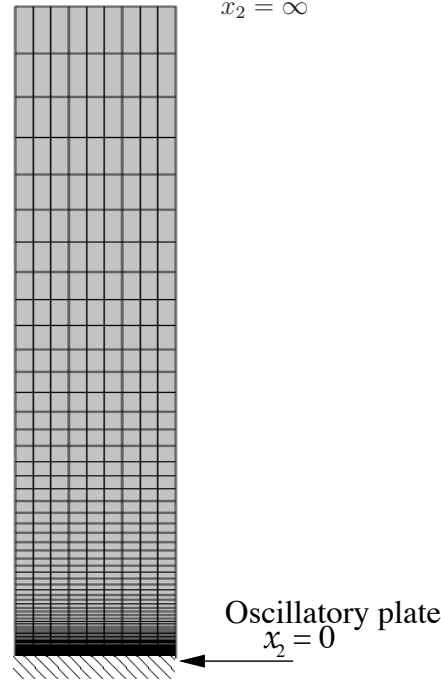


Figure 2: A side view of the numerical grid.

The height of the fluid above the plate was taken to be  $4 \text{ m} \gg \sqrt{\nu/\omega}$ . The problem is in fact a one-dimensional problem, i.e. the velocity varies only in  $x_2$  direction. Therefore in all other direction either symmetry or periodic boundary conditions are employed.

For the numerical simulations each cycle contains ten steps and the results obtain at the end of the cycle are plotted in Fig. 3. A reasonably good agreement can be found between the analytical and numerical results.

## 2.2 Oscillatory Tube

The present test case deals with the circular tube which is oscillating around its axis in the azimuthal direction in order to measure the viscosity of the fluid by the effect of shear force exerted by the fluid on its walls. A schematic view of the problem is shown in Fig. 4 where the tube with radius  $R = 8.8 \text{ mm}$  oscillates with a frequency  $f = 725 \text{ Hz}$ . The pipe oscillates in the tangential ( $\theta$ ) direction with an amplitude angle of  $\Theta_0 = 1^\circ$ . For such a configuration, the velocity of the oscillating tube is prescribed as a sine function, i.e.:

$$U_t(r = R, t) = U_0 \sin(\omega t) = R\Theta_0\omega \sin(\omega t) \quad (9)$$

where  $\omega$  is the angular frequency of the oscillations,  $r$  the radial coordinate direction and  $U_t$  the tangential velocity in the azimuthal direction  $\theta$ .

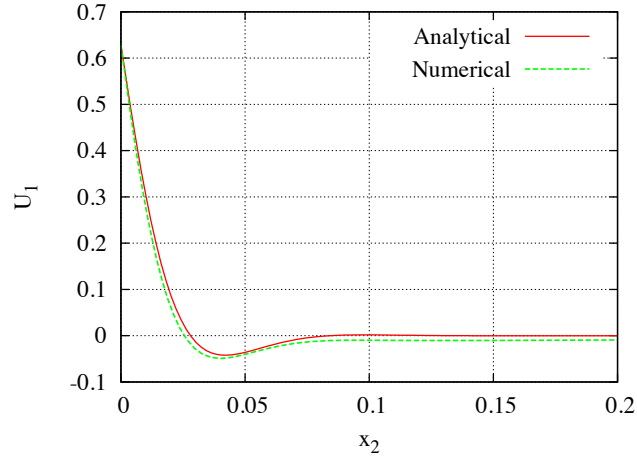


Figure 3: Comparison of periodic numerical and analytical (eq. (7)) solutions for the case of flow induced by oscillatory flat plate underneath a semi-infinite fluid at the end of a cycle.

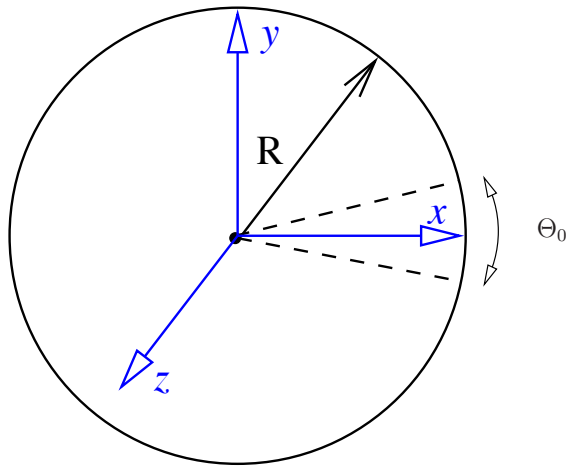


Figure 4: A schematic cross-sectional view of the oscillating around  $z$  axis.

An analytical periodic solution of the problem was derived in an earlier report [3] which has the form:

$$U_t(r, t) = \Re\{A J_1(ar)\} \quad (10)$$

$$\text{with } A = \frac{-i\omega r \Theta_0}{J_1(ar)} \text{ and } a = \sqrt{\frac{i\omega}{\nu}} \quad (11)$$

where  $\Re\{g\}$  represents the real part of function  $g$  and  $J_1(x)$  is a Bessel function of type 1 of first order.

For the numerical solution of the problem only a 2D slice (in  $x$ - $y$  plane) of the pipe is considered to be the computational domain. This is due to the fact that the problem is physically two-dimensional (however conceptually it will still be a one-dimensional problem since  $\partial/\partial\theta = 0$ ). Moreover this simplification significantly reduces the computational time. As discussed earlier, the spatial resolution requirement in a numerical computation depends on term  $\sqrt{\nu/\omega}$ , however, for the temporal resolution there is no clear indication. In fact, it is absolutely a numerical issue than a physical one. It is obvious that the numerical time step should be much smaller than physical time step (which here corresponds to frequency of oscillations, i.e.  $1/f$ ). Therefore, just to start with computations, a time step of one eighth of a cycle ( $2\pi/8$ ) was chosen.

A comparison of the computed and analytically obtained velocities for  $\nu = 1000$  cSt is shown in Fig. 5. It can be observed from the figure that agreement between the numerical computations and the analytical solution is fairly good. Since in the problem the velocities of the fluid undergo maxima (at multiple of  $\pi/2$ ) and minima (at multiple of  $\pi$ ), we must analyze and evaluate results at these two periodic locations. These two periodic locations are also referred to as imaginary and real part of the solution, respectively.

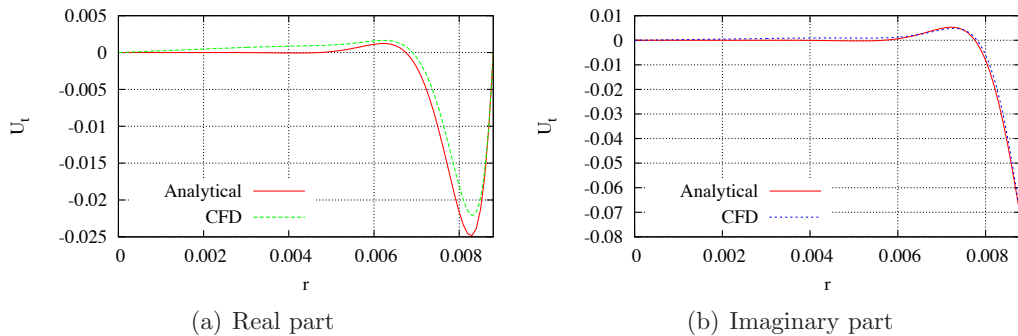


Figure 5: Periodic solution (tangential velocity profile) of the oscillating (in  $x$ - $y$  plane) circular tube with 8 time steps per cycle for  $\nu = 1000$  cSt.

For the measurement of the viscosity of a fluid, it is important to accurately estimate the forces acting on the oscillating tube. The applied average force by a fluid is directly related the shear stress exerted on the wall. Therefore, it is important to compare the

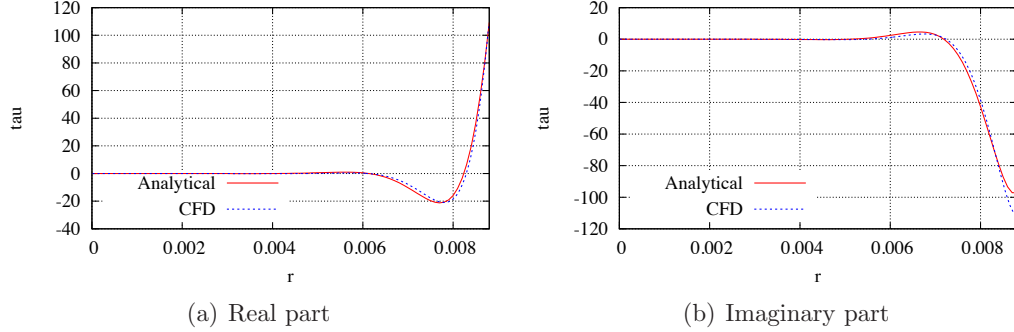


Figure 6: Periodic solution (tangential velocity profile) of the oscillating (in  $x$ - $y$  plane) circular tube with 8 time steps per cycle for  $\nu = 1000$  cSt.

numerically and analytically obtained values of the shear stress in the domain. As it can be observed from Fig. 6 that though the computed tangential velocity profiles for both real and imaginary part seem to agree well with the analytical results, the radial component of the gradient i.e.  $\partial U_t / \partial r$ , especially for the imaginary part deviate from the analytical solution in the region close to the wall. These inconsistent numerical results lead to large discrepancy in the computed *wall shear stress* for the imaginary part or at the maxima of the cycle. This is clearly indicated in Fig. 7 which shows a change in slopes of the shear rate curves for both the real and imaginary parts. The imaginary part of the shear stress (Fig. 7) shows an inversion when the viscosity is increased over a certain value. This inversion point as well as the slope of the shear stress curve is not accurately predicted with the present temporal resolution.

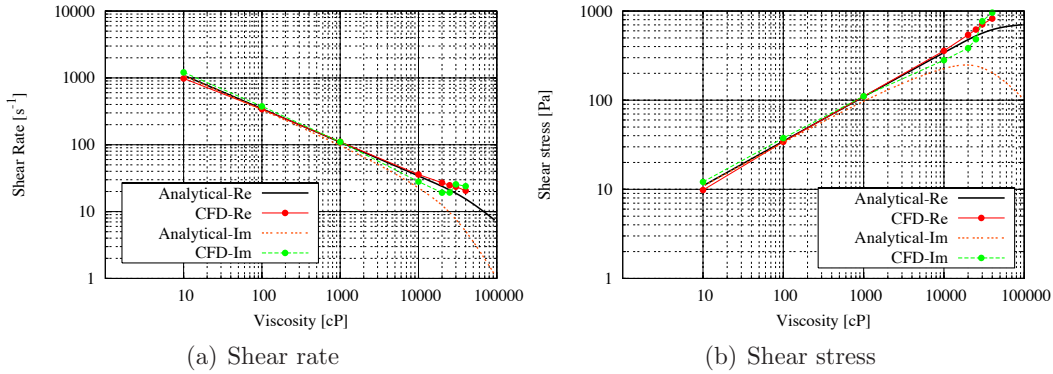


Figure 7: Numerically predicted shear rates and stresses as a function of dynamic viscosity with 8 time steps per cycle.

The reason for this disagreement was that the phase lag between the oscillating wall and fluid adjacent to it could not be well captured due to accuracy of the time marching scheme. Though the second order Euler scheme was adopted for the time integration, *eight time steps seem to be inadequate* for accurately resolving the time response of the

fluid. Taking this consideration in account, the number of time step within a cycle were increased to  $2^n$  where  $n = 4, 5 \dots, 9$  for the case with  $\nu = 20000$  cSt. The computed values of the radial shear stress, with increased temporal resolution is plotted in Fig. 8. It can be observed from the figure that the real part of the stress is more or less same

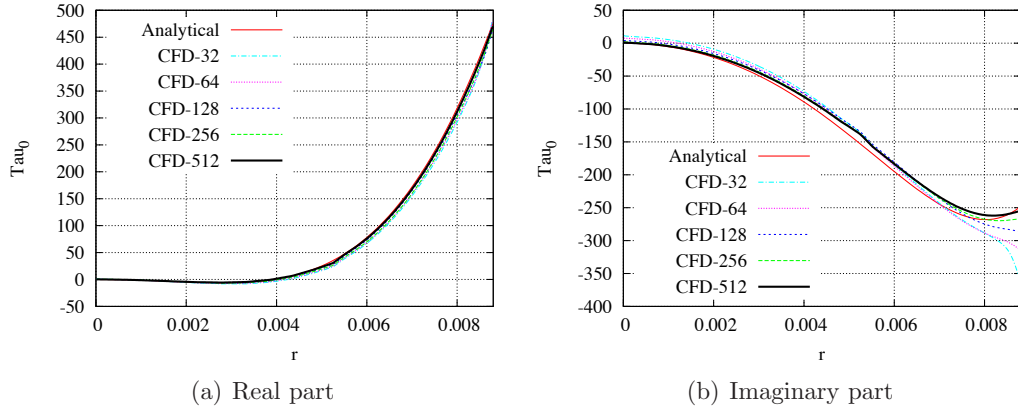


Figure 8: A comparison of the computed radial stresses with the analytical results with increasing temporal resolution as  $2^n$  where  $n = 5, 6, \dots, 9$  for  $\nu = 20000$  cSt.

for all the cases. However, the computed imaginary part indicate much improvement with increased temporal resolution. It is interesting to note that the change in sign of the second derivative of the velocity close the wall is not at all captured with temporal resolution lower than 256. This seems to be a serious limitation of a commercial CFD program like CFX where the maximum accuracy of the time integration is usually of second order. Therefore, from the present analysis it has been realized that for a reasonably accurate analysis one requires at least 256 time steps in a period especially near the point where the slope of the shear stress versus viscosity curve (Fig. 9) changes. This point is known can be termed as inversion or in-flex point. The position of the inversion in shear stress depends on the oscillation frequency and the length scale i.e. the diameter of the pipe. It has already been shown in [3] that

$$\nu_{\text{inv}} \sim R^2 f \quad (12)$$

where the subscript “inv” indicate the *inversion point* and  $R$  is the radius of the pipe and  $f$  the oscillating frequency. From the above relation one can readily conclude that the for really small frequencies and pipe diameters, the inversion point will be at relatively lower viscosity and consequently the numerical accuracy may need to be taken care even at much lower viscosities unlike in the present exercise where the inversion takes place at around 20000 cSt.

As shown earlier that with 8 numerical steps per cycle do not predict the inversion point correctly, simulations for different viscosity fluids are repeated by increasing the temporal resolution to 512. Figure 9 shows the comparison between the numerical and analytical results. The high temporal resolution computations indicate a much better agreement with the analytical results.

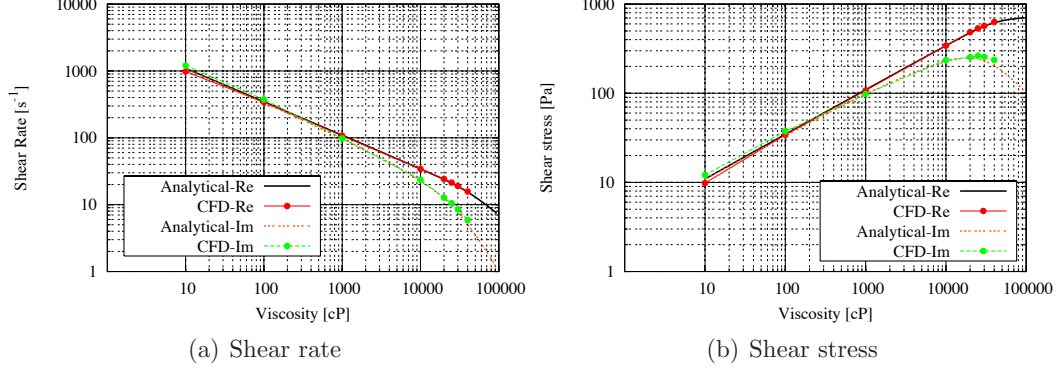


Figure 9: Numerically predicted shear rates and stresses as a function of dynamic viscosity with 512 time steps per cycle.

### 3 Conclusions

The present numerical work was carried out in order to verify the numerical simulations with the existing analytical solutions in the presence of oscillatory walls. It has been realized that for accurate numerical predictions, one requires to resolve the velocity gradients close to the wall on the basis of ratio  $\sqrt{\nu/\omega}$ . Furthermore, numerical simulations require fairly high temporal resolution in order to accurately resolve the imaginary part of the shear stress or to well define complete periodic behavior.

### References

- [1] R.B. Bird, W.E. Stewart, and E.N. Lightfoot. *Transport Phenomena*. John Wiley & Sons, 1960.
- [2] F.M. White. *Viscous Fluid Flow*. McGraw Hill Inc., 2 edition, 1991.
- [3] Christof Huber. Theorie der Viscositätsmessung mit einem Coriolis-Masse-Durchfluss-Sensor. Internal report, Endress+Hauser Flowtec AG, 2006.